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REVIEW ARTICLE

Relay cropping as a sustainable approach: problems and opportunities for sustainable crop production

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Abstract Climate change, soil degradation, and depletion of natural resources are becoming the most prominent challenges for crop productivity and environmental sustainability in modern agriculture. In the scenario of conventional farming system, limited chances are available to cope with these issues. Relay cropping is a method of multiple cropping where one crop is seeded into standing second crop well before harvesting of second crop. Relay cropping may solve a number of conflicts such as inefficient use of available resources, controversies in sowing time, fertilizer application, and soil degradation. Relay cropping is a complex suite of different resource-efficient technologies, which possesses the capability to improve soil quality, to increase net return, to increase land equivalent ratio, and to control the weeds and pest infestation. The current review emphasized relay cropping as a tool for crop diversification and environmental sustainability with special focus on soil. Briefly, benefits, constraints, and opportunities of relay cropping keeping the goals of higher crop productivity and sustainability have also been discussed in this review. The research and knowledge gap in relay cropping was also highlighted in order to guide the further studies in future.

Keywords Crop productivity · Environmental sustainability · Intercropping · Resource use efficiency · Soil health

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Introduction

Ensuring global food security with changing environment and shrinking natural resources are the major challenges in the present era (Hussain et al. 2015). Population density has risen rapidly over the last few decades, and supply food to such a growing population is a challenge all over the world, and resultantly, agriculture people should optimize the use of resources especially soil, and the soil management is very important for healthy and sustainable food production (Keesstra et al. 2016a). Sustainable use of natural resources in the face of high population density is critically important and consequently food insecurity is overwhelming for specially developing countries (Araya et al., 2010; Musinguzi et al., 2015; Nyssen et al., 2015; Tsozué et al. 2015). The increasing population pressure on land and water resources leads to the degradation of these resources, which often results in the loss of productive capacity and food insecurity as a consequence of soil degradation and the increase in soil erosion (Borrelli et al. 2015; Gessesse et al. 2015; Ochoa-Cueva et al., 2015).

Farmers have responded to the pressure on the use of land and the associated decline in productivity by expanding into the fragile bottomlands and steep slopes (Ligonja and Shrestha, 2015; Nanko et al. 2015). This situation has led to the excessive exploitation of natural resources and increased

soil losses (Novara et al. 2015; Keesstra et al. 2014). Numerous factors like intensive tillage (Lieskovský and Kenderessy, 2014; de Moraes Sa et al. 2015; Laudicina et al. 2015), soil erosion, compaction and organic matter depletion (Cerda et al. 2010;), indiscriminate and overuse of pesticides and chemical fertilizers (Cerda et al. 2009; Hussain et al. 2015), and reduced vegetation cover (Novara et al. 2013a; Keesstra et al. 2015) are reported to trigger the soil degradation. Intensive agriculture-based practices affect many ecosystem services essential to humans and the environment such as nutrient cycling; organic matter decomposition; regulation of C and N fluxes; decontamination of the environment; and control of pests and diseases for humans, animals, and plants (Mekonnen et al., 2015; Brevik et al. 2015). Emissions of different greenhouse gases from agricultural soils and leaching of nutrients into underground water table are deteriorating the natural ecosystem (Parras-Alcántara et al. 2013; Novara et al. 2013a).

In order to reduce the soil degradation and enhance the ecosystem sustainability, various strategies such as use of cover crops and buffer strips (Mutch and Martin 1998; Adimassu et al. 2014; Novara et al. 2013a), no-till or minimum tillage practices (Shah et al. 2016), use of geotextiles (Gimenez-Morera et al. 2010), organic amendments like biochar (Weyers, 2014), and mulching (Cerdà et al. 2016; Prosdocimi et al. 2016) are reported elsewhere. Relay cropping is one of most reliable and applicable practices including a complex suite of different resource-efficient technologies.

Relay cropping is a method of multiple cropping where one crop is seeded into standing second crop well before harvesting of second crop (Queen et al. 2009). Relay cropping is a sustainable approach that optimizes system productivity and compensates yield of two crops at a time and can solve time contravene among sowing of different crops. It possesses the capability to improve soil quality, to increase net return and land equivalent ratio, and to control the weeds and pest infestation, thereby decreasing chemical pest control measures (Jabbar et al. 2011; Bandyopadhyay et al. 2016).

Relay cropping facilitates the farmers to cultivate two crops in 1 year especially in those areas/cropping systems where growing season is shrinking for sequential farming due to climate change (Akanvou et al. 2002; Reda et al. 2005; Zhang et al. 2007a; Jabbar et al. 2010). Other environmental benefits associated with relay cropping include improved soil, air, and water quality by reducing the leaching, emanations, and eutrophication of nutrient compounds (Schepers et al. 2005). Relay cropping has still been recognized, especially by smallholder farmers, because of its potential to increase land use efficiency and reduce fertilizer consumption (Ayisi et al. 1997), enhance crop yield and nutrient accumulation, and improve biological activities (Ghosh et al. 2006). Relay cropping is proposed as a beneficial tool that results in better utilization of residual soil moisture from previous crops and reduces cost of production per unit area (Jabbar et al. 2005; Zhang et al., 2008a). However, majority of the studies pertaining to relay cropping have also highlighted several constraints including low germination problems and interspecific competition particularly in developing countries. The objectives of the current review article are (i) to identify the potential benefits of relay cropping in agro-ecosystem (ii) to examine the role of relay cropping in reducing soil degradation and enhancing sustainability, (iii) to discuss the scope of relay cropping in plant protection and plant nutrient economy on sustainable basis, and (iv) to highlight the research and knowledge gap in relay cropping, in order to guide further studies in the future.

Materials and methods

In the present review, contemporary synthesis of the existing data was conducted in order to examine the potential benefits of relay cropping in agro-ecosystem and how relay cropping can be helpful in minimizing the challenges of modern agriculture. This review includes critical analysis of 128 scientific journal articles, 5 book chapters, and 9 other publications (online reports, conference proceedings, and thesis). Most of the publications were downloaded from the Web of Science (WoS), using Science Citation Index.

Relay cropping in the context of crop productivity, net economic returns, land use efficiency, soil quality, nutrient economy, and pest control was discussed and after literature review, it was realized that the intensive-based agriculture practices (e.g., intensive tillage, indiscriminate and overuse of pesticides, and chemical fertilizers) are severely affecting the ecosystem services essential to humans and the environment such as nutrient cycling, organic matter decomposition, and regulation of carbon and nitrogen fluxes. In view of above, an attempt was tried to link our hypothetical concepts to previous evidences and provided a comprehensive assessment regarding the role of relay cropping in enhancing ecosystem sustainability and reducing soil degradation. Further, the scope of relay cropping in future agriculture was also discussed and specific key research and knowledge gap in this to guide the further studies was also highlighted.

Results and discussion

Potential benefits of relay cropping

Increase in crop yield and net return

Relay cropping is an efficient crop production system that sustains the income of small farmers through increasing crop productivity and net return per unit area (Yildirim and Guvenc 2005). Yield benefits from relay cropping are positively linked

with efficient utilization of natural resources. Additional income and yield advantage can be obtained by raising cotton (*Gossypium hirsutum* L.) in wider rows followed by any relay crop than a sole cotton crop (Rao 1991). In another study, cotton yield was unaffected in relay cropping systems; nevertheless, considerably higher yield of wheat (*Triticum aestivum* L.) (69.37%) and barley (22.84%) was achieved from relaying of wheat and barley (*Hordium vulgare* L.) in cotton compared with conventional wheat and barley planting after the harvest of cotton (Khan and Khaliq 2005). Furthermore, Reda et al. (2005) observed that relay cropping of *Sesbania sesban* L. considerably increased the yield of non-fertilized sorghum (*Sorgum bicolor* L.) by 67% during the first year and by 37% during the second year of experiment as compared with the sorghum crop alone. The difference in yield increment was due to the prolonged dry spell during the second year of the experiment. Some studies have reported reduced yield of succeeding crop when relayed in primary/standing crop, but such reductions might be due to the early-stage competition faced by the second crop during its seedling stage. For instance, in wheat–cotton relay cropping system, fully developed wheat plants compete for light, space with cotton seedlings, and reduced canopy development of cotton (Zhang et al. 2007a). Decrease in yield might also be due to over shading effects (Akanvou et al. 2002), malfunctioning of roots and interference among interspecific roots (Gill et al. 2009), substantially different resource use potential, and significantly varying growth behavior (Ahmad et al. 2007). Contrarily, no significant yield difference of soybean (*Glycine max* L.) was observed when planted as relay crop in wheat and/or as monocrop soybean (Wallace et al. 1992). The feasibility and interaction of different gram-based relay cropping systems in rainfed areas was examined in a study, and it was found that gram yield was considerably reduced by 3.54%, 4.12%, 6.06%, 7.66%, and 29.86%, when relay intercropped with mung bean (*Vigna radiata* L.), lentil (*Lens culinaris* L.), linseed (*Linum usitatissimum* L.), wheat (*T. aestivum* L.), and brassica (*Brassica napus* L.), respectively (Nazir 1992). Moreover, such loss in gram yield was compensated later with the additional yield advantage from mung bean (470 kg ha^{-1}), lentil (646 kg ha^{-1}), linseed (606 kg ha^{-1}), wheat (227 kg ha^{-1}), and brassica (903 kg ha^{-1}) and suggested that the use of such gram-based relay cropping in rainfed areas is an effective way of increasing the income potential of small land hold farmers (Nazir 1992). Recently, Yamane et al. (2016) documented that the annual soybean production in relay intercropping was greater that double of the cropping system.

Rice (*Oryza sativa* L)-based relay cropping systems were found more productive, economically viable, and remunerative than the rice/fallow cropping system (Jabbar et al. 2005). They further reported (based on total rice yield equivalents) that forage legume relay cropping systems such as rice/fenugreek (*Trigonella foenum-graecum* L.) (7.97 and 9.27 t ha^{-1}), rice/chickpea (*Cicer arietinum* L.) (6.96 and 7.60 t ha^{-1}), rice/lentil (*Lens culinaris* L.) (6.98 and 7.47 t ha^{-1}), and rice/Egyptian clover (*Trifolium alexandrinum* L.) (7.13 and 8.13 t ha^{-1}) were found superior to rice/non-legume relay cropping systems such as rice/wheat (5.66 and 6.52 t ha^{-1}), rice/barley (6.45 and 7.69 t ha^{-1}), rice/brassica (5.15 and 5.87 t ha^{-1}), rice/sunflower (*Helianthus annuus* L.) (6.39 and 8.64 t ha^{-1}), and rice/forage maize (*Zea mays* L.) (4.64 and 5.03 t ha^{-1}) under zero and conventional tillage system, respectively (Jabbar et al. 2005). Likewise, a rice–pea (*Pisum sativum* L.) relay cropping system gave higher net returns than did a sole rice crop (Agarwal et al. 1986). Relay cropping of pigeon pea (*Cajanus cajan* L.) did not reduce the groundnut (*Arachis hypogaea* L.) yield and resulted in additional income (Chodavadia et al. 2013). Relay cropping of lathyrus (*Lathyrus sativus* L.) or maize fodder (*Z. mays* L.) crop in rice gave higher benefit cost ratio (4.00) (Alam and Costa 1988). Likewise, a wheat–maize relay cropping system has been reported to increase grain yield (161.5% to 268.2%) than did sole wheat crop (Zhang et al. 2007b). Relay cropping was proposed as a more profitable system for farmers with reduced risk of crop failure or problems linked with double crop system (Nelson et al. 2010). They evaluated the relay cropping of wheat in soybean and suggested that utilization of high-yielding wheat cultivars and coated soybean should be used to maximize the gross profit (Nelson et al. 2010).

Increase land use efficiency

Land use efficiency can be assessed by measuring land equivalent ratio (LER). The LER is the ratio of the land area used under sole cropping to the land area used under relay cropping/intercropping, needed to give equal amounts of yield at the same management level. Relay cropping increases the land use efficiency by increasing LER. For instance, Hussein (2005) found that relay cropping of cotton in wheat increased the land use efficiency by 81% –213% depending upon the sowing time of cotton. Land productivity and net return was also enhanced in relay cropping systems as compared with monoculture (Zhang et al. 2007a, b). A number of studies have confirmed the benefits of relay cropping in increasing land use efficiency and LER, when compared with monoculture (Table 1). Akram et al. (2004) and Dua et al. (2007) recorded higher LER of berseem (*T. alexandrinum L.*) (73%) and potato (*Solanum tuberosum* L.) (79%) when sown in sunflower and wheat respectively as compared with respective monocrops. Yamane et al. (2016) reported that LER values for relay intercropping of soybean were near 2.0, which were considerably higher than the double cropping system (LER 1.62). The values of monocropping for two cultivars were near 1.0 (Yamane et al. 2016). Conclusively, the relay cropping system is an example of temporal diversification usually intended to increase the land use efficiency and

Table 1 Land equivalent ratio and relay cropping system

Relay crop	Sole crop	LER of relay crop[a]	Reference
Berseem	Sunflower	1.73	Akram et al. (2004)
Potato	Wheat	1.79	Dua et al. (2007)
Cotton	Wheat	1.39	Zhang et al. (2007a)

LER land equivalent ratio

[a] LER was the ratio of the area under sole cropping to the area under relay intercropping needed to give equal amounts of yield at the same management level

yield per unit of the cropping area while enhancing ecosystem services.

Facilitate timely sowing of crops

Rice–wheat and cotton–wheat cropping systems are the two major systems of developing countries in semiarid to subhumid areas. On the other hand, maize and soybean-based cropping systems are the major systems in temperate regions. In recent years, these systems have suffered with delayed planting of succeeding crop (Gangwar and Prasad 2005), which results in tremendous yield loss of succeeding crop (Saeed et al. 1999). Nevertheless, relay cropping has potential to increase the production level by increasing yield and yield attributes of a crop due to timely sowing of second crop. Wheat relayed in cotton gave a significantly higher grain yield (2964 kg ha^{-1}) against the wheat crop sown after the harvest of cotton (1750 kg ha^{-1}) (Khan and Khaliq 2005). Higher grain yield of wheat relayed in cotton was attributed primarily to higher plant density (6%), more fertile tillers $plant^{-1}$ (21%), grain $spike^{-1}$ (18.6%), and higher 1000-grain weight (13.2%) as compared with the wheat crop sown after the harvest of the cotton. The wheat crop relayed into cotton experienced a favorable temperature for germination, crown root development, and subsequent developmental stages. The crop sown after harvest of cotton not only faced a low temperature but also received a smaller growth duration period (about 1.5 month). Andrew et al. (1990) also found similar results as was of Khan and Khaliq (2005) that wheat crop sown after the harvest of cotton crop not only experienced unfavorable temperature regimes but also short growth duration, which reduced wheat yield. The high temperature at post-anthesis/grain filling stage of late-sown wheat led to poor development of food storage organs and reduced seed weight (Anwar et al. 2011). Each growth stage during crop cycle requires specific thermal heat units; nonetheless, delayed sown crops remain unable to accumulate sufficient heat units, which ultimately reduce yield and yield-related traits (Noworolnik and Pecio, 1990; Sajjad et al. 2014). Relay cropping allows cotton to grow in rotation with winter wheat when there is no difference between heat resource availability and heat requirement of cotton if sown after harvest of wheat (Zhang et al. 2007a). It was suggested the relay cropping of cotton and maize respectively in standing wheat crop in order to ensure timely sowing of the second crop (Zhang et al. 2007a & Yan et al. 2010). Relay cropping compensates reduction in plant population due to late sowing by giving an opportunity to farmers to sow the intended crop in time. Nasrullah et al. (2010) found that wheat yield was reduced by 1.0%–1.5% per day delay in sowing beyond the 20th of November. However, preliminary research regarding relay intercropping revealed that wheat recorded substantially higher grain yield and biomass when its surface was seeded in standing cotton than delayed sowing after cotton harvest (AARI, 1992). A negative correlation was also found between delayed wheat sowing and different yield components (Jan et al. 2000); however, relay cropping possesses a great scope by allowing timely sowing of succeeding crop and facilitates the crop to complete its growing degree days and furnish more yield. According to El-Hawary (2009), relay cropping permits summer crops to be planted in standing winter crop to avoid the delayed planting and yields more net return than does a sequential double cropping system. In a study, comparisons were made between double cropping and relay intercropping of black soybean and found that the yield of second soybean crop in double cropping was prone to decrease due to late sowing; nevertheless, such yield losses were not observed for relay crop because of timely sowing (Yamane et al. 2016).

Improve soil quality

Modern agriculture with highly mechanized and chemical farming has led to various problems that are deteriorating the soil quality (Cerdà et al. 2009; Bruun et al. 2015; de Oliveira et al. 2015). Relay cropping showed impressive results not only in preventing the soil from further deterioration or degradation but also in improving soil health. Inclusion of relay crops may reduce the nutrient losses from the soil leading to less negative impact on environment. Relay crop has a scavenging effect on residual soil nutrients and uses these available nutrients which would be lost otherwise. For instance, planting of wheat in standing seed corn scavenged 130 kg N ha^{-1} from the root zone and produced an extra 2-t residue ha^{-1} (Schepers et al. 2005). They further suggested that wheat relay crop reduced the nitrate leaching, because of efficient utilization of precipitation (that might result in nitrate leaching) by the young wheat, and therefore, the nitrate in the root zone was utilized by the scavenger crop. Although the concept of wheat planting as a scavenger crop after seed corn production has environmental merits, considerable uncertainty is associated with the scheme in terms of contributing to producer profitability. This is because adequate N must be supplied by the soil or made available from fertilizer to

grow the crop and develop the grain to be harvested. Moreover, the N carry-over after seed corn production may vary with weather conditions and previous management practices (Schepers et al. 2005). The planting of red clover (*T. pratense* L.) or white clover (*T. repens* L.) as relay intercrop significantly reduced the N losses due to leaching during winter compared with the absence of clover (Askegaard and Eriksen 2008). Additionally, it was also suggested that farmers can manage the N content and its distribution into the soil leading to reduced fertilizer wastage and N pollution of surface and groundwater (Novara et al. 2013a). Tillage and herbicides prevented the soil development and enhanced the erosion rates; however, the management practices that trigger the use of legume crops, straw mulch, and plant cover may contribute to an increase in soil structural stability and reduce/eliminate the soil and water losses (Keesstra et al. 2016b). Relay cropping of legume can improve the soil nutrient status, particularly N through biological nitrogen fixation and nutrient recycling (Hartemink et al. 2000, Tian et al. 2000). Relay cropping possesses the ability that may create a healthy soil microenvironment (Dumanski et al. 2006). It involves the minimum or no tillage techniques leaving a considerable amount of crop residues on soil surface which concomitantly increase the soil organic matter. Different studies indicated that adoption of zero tillage in relay cropping systems provides considerable improvement in soil quality as compared with conservation tillage. Zero tillage typically improved the soil quality by regulating soil fertility, soil structure, and soil biological properties (Tanveer et al. 2014; Ashraf et al. 2014). The cropping system comprised of exhaustive crops (e.g., rice–wheat or cotton–wheat system) typically having low soil organic carbon (SOC) (Duxbury et al. 2000; Hulugalle and Scott 2008), while zero tillage as a part of relay cropping can enhance the SOC in the surface layers and lead to higher stability of soil aggregates. Relay cropping reduces cultivation costs in a cropping system by reducing the number of autumn tillage, which also affects soil fertility. Relay cropping reduced soil compaction and increased humus contents thus improving soil structure (Snapp et al. 2005).

Residue burning is a common practice to get rid of residues and facilitate land preparation; however, it reduces the biological functions of organic matter, soil aggregation, and carbon sequestration and directly through a loss of organic matter input (Giller et al. 1997). Fire can be the one responsible for the reduction of soil organic matter (Novara et al. 2013b) but also to the loss of a large amount of charcoal due to the high erosion rates measured after forest fires (Keesstra et al. 2016; Pereira et al. 2016). Moreover, residue burning detrimentally affects the global atmospheric chemistry by adding gaseous and particulate pollutants to the atmosphere (Xinghua et al. 2007). Relay cropping provides an opportunity to avoid the residue burning, for instance, Punyalue et al. (2015) suggested that maize + legume relay cropping systems are promising to reduce the burning in highlands and to improve soil nutrient status and maize yield. The maize grain yield was enhanced by 24–53% in the maize–legume relay cropping system compared with monocropping of maize and indicated that biological N_2 fixation in legumes most likely contributed in enhancing the total aboveground nitrogen and grain yield as well.

Efficient use of natural resources

Optimization and efficient use of natural resources might be achieved by employing relay cropping system. For small land holdings, resource-based conservation technology is the best weapon to sustain the production potential of their farms and relay intercropping is very attractive in that context (Ndah et al. 2015). It was also noted that cover crops as relay crops in any commercial crop may increase resource use efficiency and biomass production (Scopel et al. 2013). In rainfed areas, rainfall is the only source of irrigation. During excessive rainfall periods, excessive water can be used to produce more yields per unit area by increasing water use efficiency. Relay cropping helps to intensify farm area to use more available water for a limited time (Ali and Sarker 2013). Similarly, under drought conditions, employing drought-resistant crops as relay crops not only reduces exposure of land area to high temperature but also provides soil cover. Furthermore, yields of relay-cropped soybean were 58–83% of full-season monocrop, while they were higher than those for double-cropped soybeans (Gesch et al. 2014). Although the sowing time was the same, the lower yield of relay-cropped soybean was likely due to the competition with camelina (*Camelina sativa* L.) crop for available growth resources during the initial growth stages. Similar yield reductions were also reported by Wallace et al. (1992) in soybean interseeded with wheat. Moreover, similar to Gesch et al. (2014), the higher yield of the relayed cropped soybean (*Glycine max* L.) than of the double-crop soybean were also found for wheat–soybean-based dual crop systems in Missouri (Nelson et al. 2011) and Kansas (Duncan and Schapaugh, 1997). These authors have attributed the higher yield of relay-cropped soybean to earlier sowing date than of the double-crop soybean. Minimum soybean seed yield was recorded for the double-seeding after camelina. Soil water availability in the months of June, July, and August is critical for the growth of double and relay crops (Berti et al. 2015). Gesch et al. (2014) as well as Berti et al. (2015) suggested that relay cropping of soybean (*Glycine max* L.) in camelina (*Camelina sativa* L.) crop can extract the soil water in June and early July, and its root system is developed well before soil water becomes limiting by mid-July and August in most years. After the physiological maturity stage of camelina (*Camelina sativa* L.) (sometime by late June), it does not compete for soil water with the relay crop. High radiation use efficiency and/or higher resource use

efficiency of relay cropping could be attributed to the better distribution of the plants in space and probably reduced competition between the two crops.

The effectiveness and profitability of relay cropping is closely associated with the performance of crops under field conditions, which depends on the availability of resources and time. For instance, Martens et al. (2001) concluded that the availability of resources during the reproductive/late stage of the first crop and early growth stage of the relay crop is more important in examining crop performance. Furthermore, it was also suggested that relay intercropping of black gram in flood-affected rice (*O. sativa* L. rainy-season variety Aman) could be a resource-efficient technique that not only improves net return per unit area but also utilizes extra water in rice (Maniruzzaman and Miah 1989). It was recommended that growing off-season vegetables in standing crop may provide additional source of income (Tuti et al. 2012). Similarly, for winter crops, with the advancement in sowing or transplanting through relay intercropping, one good off-season crop can be taken during the spring season and in this manner, four crops can be grown in a year successfully. It was further suggested that the relay cropping of corn after flue-cured tobacco apex excision could become a profitable cropping system in Yunnan and other regions with similar environmental conditions (Zhou et al. 2014). Besides all these, a relay cropping system could also be an option to produce biofuels, food, and biomass feedstock in a single season on the same land without sacrificing food security.

Although different studies have indicated that the relay cropping increases the land use and net return, if further studies should be conducted to improve our understanding pertaining to possible benefits of relay cropping, research should include quantitative assessments of increment in land equivalent ratio, gross monetary return, provision of time for timely sowing, utilization of available resources, and improvement in soil quality.

Nutrient economy

In intensive cropping system, management of nutrient is of significant importance. In relay cropping, the nutrient availability to succeeding crop or the influence of succeeding crop on nutrient requirement of the primary crop emphasizes the need of studying nutrient economy. In a study, it was examined that the contribution of relayed legume cover crops on N dynamics and grain yield in an organic wheat/relay intercropped legume/maize production system and observed that the enhanced N nutrition due to relayed legume crop increased the maize yield by 30% and N content in aerial plant parts of maize by 55% compared with control (Amossé et al., 2014). Red clover accumulated dry matter up to 1800 kg ha^{-1} when relay was cropped in winter wheat in southern Manitoba conditions (Martens et al. 2001). Relay crops utilize the nutrients more than monoculture crops do. Willey (1990) proposed that intercrops may enhance the ability of mixed crops to capture leached nutrients from the soil more than monoculture crops could. Intercropping gives yield benefit, but that could be due to extra inputs of fertilizer to the intercropped crops (provided that both are exhaustive crops); however, this is not in the case of legumes when they were relayed with cereal legume and fixed N biologically. Biological nitrogen fixation by legume relay crop improves the fertility status of the soil (Fujita et al. 1992). For instance, Jabbar et al. (2005) found that soil N contents in rice–wheat relay cropping systems were reduced by 2.4–9.8%, while they increased by 2.5–9.7% in rice–legume-based relay cropping systems compared with the control. In another study, Jabbar et al. (2011) examined the residual soil fertility as influenced by diverse rice-based relay cropping systems and found that Egyptian clover-oriented relay cropping increased more residual N (53.6%) and organic matter (38.2%) as compared with lentil (*Lens culinaris* L.)-oriented inter/relay cropping systems, while wheat-oriented inter/relay cropping systems either decreased or did not add N and organic matter in the soil. It has been elucidated through various investigations that nutrient use efficiency was relatively higher in intercrops than in sole crops due to high N uptake and biomass production (Zhang et al., 2008b). The same amount of N was also taken in mono-rice and rice intercropped with grain legumes, but there was significantly less dry matter production in intercropped rice (Aggarwal et al. 1992). Contrarily, it was pointed out that relay intercropping of legumes in wheat can provide reactive N to the soil–plant system through symbiotic N fixation (Amossé et al., 2014). Such approaches can be useful in increasing soil fertility on sustainable basis particularly in N-deficient cropping systems (Amossé et al., 2014). Inclusion of relay legumes in organic grain production system has been found to be a profitable and ecologically sustainable technique (Berry et al. 2002; Casagrande et al. 2009; David et al. 2005; Mazzoncini et al. 2010; Gaudin et al. 2014). Relay cropping of red clover in winter wheat had no negative effects on wheat grain yield and some other studies also investigated the effect of inorganic N availability on the behavior of various spring barley and winter wheat–legume relay cropping systems and reported similar effects on N economy (Queen et al. 2009; Bergkvist et al. 2011). Decrease in N rates can maximize the economic returns from wheat–red clover relay cropping and provide higher partial profits using less N inputs than single crops do (Gaudin et al. 2014).

In relay cropping systems, zero tillage facilitates crop residue retention, which ultimately affects the N availability especially in early stages of new crop. Incorporation of legumes in soil and use of clover or legume-oriented relay cropping systems improve the nutrient status of soil and organic matter in soil. Relay-cropped jack bean provided additional N input through biological N_2 fixation (Pansak et al. 2008). Furthermore, pigeon

pea was the most suitable legume crop in improving soil fertility when used as relay crop in upland rice (Akanvou et al. 2002). Extracellular enzymatic activities in soil were due to the increase in vegetation cover, which ultimately affected the degradation processes (Tejada and Benitez 2014). Interseeding of clovers provides a number of benefits like weed suppression and soil cover to reduce erosion and provides plenty of N to succeeding crop (Vyn et al. 1999; Zhao et al. 2013). Planting of the appropriate cover crops like *T. subterraneum*, *Festuca rubra*, *F. ovina*, and *Lolium perenne* between the rows of vineyard effectively trapped the water and decreased soil erosion losses (Novara et al., 2011). The residual nutrient availability mainly rely upon the type of relay cropping system used, e.g., lentil and wheat-oriented relay cropping systems have no significant effects on residual nutrient status in soil while clover-oriented systems increased markedly soil organic matter (13.33%) and residual soil N (9.52%) (Jabbar et al. 2011). Sesbania–maize relay cropping systems as a strategy to improve soil fertility and food production (Phiri et al., 1999). Relay cropping is considered the best approach to enhance soil organic matter and soil nutrient recovery by the addition of legume crops in relay cropping systems. The increment in yield might be ascribed to addition of more organic matter in soil and availability of biologically fixed N to preceding crops (Bhattacharyya et al. 2009). Shah et al. (2003) also tested mung bean–wheat and lentil–maize cropping systems, and they ascribed the increase in soil organic matter and soil N to the inclusion of legumes in these relay cropping systems.

Plant protection

Pest control

Relay cropping possesses a great potential to suppress pest attack either by increasing the no. of predators or by disturbing the pest population dynamics (Figs. 1 and 2; Mensah and Sequeira 2004; Ma et al. 2006). In a study, it was found that wheat as winter relay crop not only repressed cotton aphid population but also served as reservoir for predators of cotton pest (Men et al. 2004). Relay cropping was also found effective in reducing *S. hermonthica* infestation (27.4–38.7%) in sorghum (Reda et al. 2005). On the other hand, Phoofolo et al. (2010) indicated that the abundance and species composition of lady beetles between relayed/intercropped (wheat) cotton, alfalfa, and monoculture sorghum were similar. The aphid population was significantly reduced by 46.7% in the cowpea + sorghum intercrop, but it was 7% higher in the cowpea + greengram intercrop as compared with mono-cowpea crop (Nampala et al. 2002).

Relay cropping is actually a prototype system that makes a predator reservoir prior to advent of key pests of primary crops. However, the extent of pest control generally depends on crop selection and behavioral response between pest and predators (Parajulee and Slosser, 1999). Crop selection is the key for relay cropping, especially when that crop is intended to act as reservoir for predators. Therefore, a crop having the ability to attract natural enemies should be preferred. Controversies are there on considering relay cropping as potential biological, physical, and ecological method for pest management. However, according to Parajulee and Slosser (1999), canola–sorghum relay cropping systems increased the total predators by 18.69 and 9.39% as compared with canola alone and forage sorghum, respectively, on the other hand, the total no. of aphids was reduced by 8.9 and 4.2% due to canola–sorghum relay cropping as compared with fall canola and forage sorghum alone, respectively. However, there was no difference in the population of lady bird beetle between cotton–wheat relay cropping systems and monoculture (Slosser et al. 2000). Therefore, selection of a suitable crop in enhancing biological control of pests is important in relay cropping systems. Relay cropping of cauliflower and lettuce to manage the attack of cabbage maggot species (*Delia radicum*, *D. platura*, and *D. florilega*) and cauliflower yield loss (Parsons et al. 2007). They further concluded that first generations of these cabbage maggot species were the most destructive to newly transplanted cauliflower crop and that the relay cropping significantly reduced the no. of *D. radicum* (70.1%) and *D. platura/D. florilega* (28.68%) eggs as compared with mono-cauliflower crop. Several other researchers also found the successful results of relay cropping for pest management; for instance, efficient pest control was observed for relay cropping of corn in bean (Altieri et al., 1978). Relay cropping of cabbage in tomato decreased the incidence of pests on cabbage crop (Xu et al. 2010). They further pointed out that shelter provided by tomato precluded adult insects to lay eggs in cabbage.

Weed control

Weeds are the most dramatic biological entities that interfere with crop performance and consequently cause substantial yield losses by competing for light, nutrients, and water (Bajwa et al. 2014). Moreover, the allelopathic effects of weeds have also been well-reported for several field crops (Hamidi and Ghadiri, 2011). Weeds may constitute a physical barrier which can affect the soil seed bank (Doisy et al. 2012). Nonetheless, weeds might be controlled by producing unfavorable conditions for them. The inclusion of forage legumes in winter cereals was anticipated to provide soil cover by the time of cereal harvest and concurrently enables successful weed control (Sjursen et al. 2011). Relay intercropping of forage legumes in organic winter wheat appeared to be efficient at tackling the weed issue, and the presence of a forage cover crop significantly decreased the density and biomass of spring-germinating annual weeds (avg. 35.2%) during the



Fig. 1 Pictorial view of **a** cotton as relay crop in wheat and **b** maize as relay crop in wheat. **a** and **b** has been adopted from Zhang et al. (2007a) and Yan et al. (2010), respectively

relay intercropping period (Amossé et al. 2013). Numerous other studies have also demonstrated considerable weed control due to the adoption of relay cropping (Table 2).

The potential of relay forage legumes for reducing weed infestation varied among different species, and such variations were dependent upon their morphological, phenological, and physiological characteristics (Blackshaw et al. 2010; Uchino et al. 2011). However, an array of growth traits and branching patterns (aboveground and belowground) may also allow a legume to fit into a cropping system as a relay crop. In addition, the total weed population may also determine the extent of weed control by relay cropping systems. Subpopulations of weeds can react differently in the presence of relay legumes as cover crops (Hiltbrunner et al. 2007). Different relay crop treatments reduced the germination of spring germinating weeds (50.35%) and summer/autumn germinating weeds (10.06%) (Hiltbrunner et al. 2007). Relay cropping also decreased the dry matter accumulation of dicotyledonous weeds and monocotyledonous weeds by 68.5 and 39.36%, respectively. The presence of relay crop prior to harvesting of previous crop may limit the weed growth (Amossé et al. 2013). Higher competition for resources, especially light, was the reason for less weed emergence and higher mortality in relay cropping systems (Kruidhof et al. 2008). Relay mucuna crop in maize severely hampered the weed growth; therefore, weed biomass in relay-cropped mucuna treatments was 89% lower than in non-mucuna plots (Correia et al. 2014).

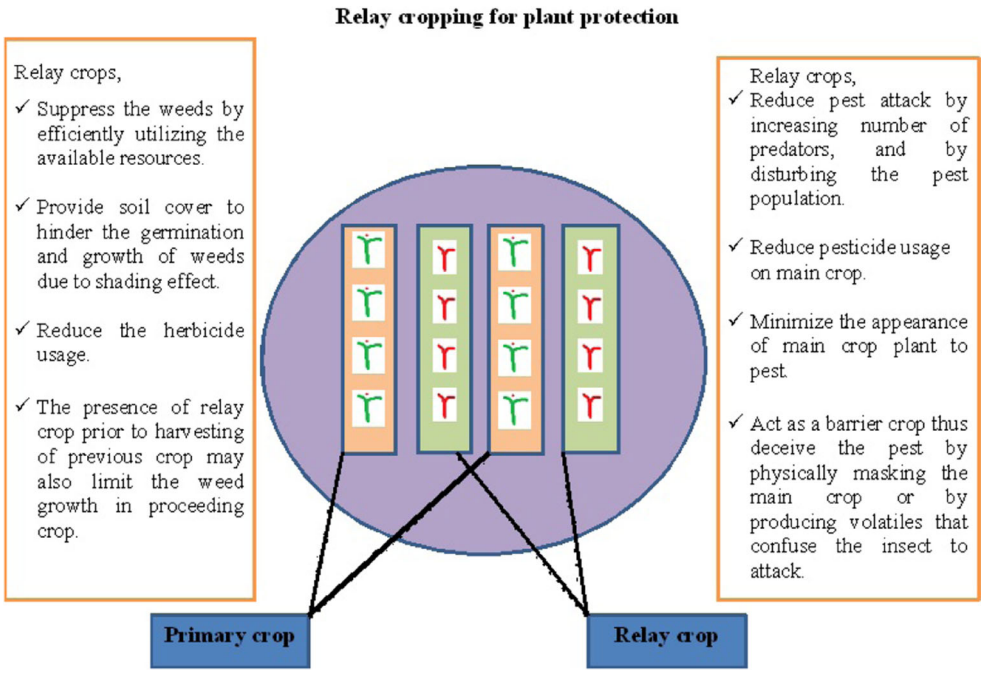


Fig. 2 Role of relay cropping in pest control

Table 2 Potential of relay cropping in reducing weed infestation

Relay intercrops	Sole crop	Weed type	Weed control	Reference
Legume crops	Winter wheat	Spring germinating annual weeds	35.2%	Amossé et al. (2013)
Sesbania sesban L. and *Cajanus cajan* L.	Sorghum	*Striga hermonthica*	62.1%	Reda et al. (2005)
Canola and sunflower	Wheat	Spring germinating annual weeds	34.5%	Qayyum et al. (2013)

Optimization, adoption, and scope of relay cropping systems

Modeling to predict and optimize system productivity

Computer models of individual crops are being applied to evaluate the goodness of particular crop combinations in relay cropping systems (Zhang et al., 2008b). Economic models could be more effective in estimating the yield compensation feature of relay cropping (Saeed et al. 1999; Zhang et al., 2008a). The non-linear behavior of relay cropping systems cannot be accounted for by simply studying or modeling component crops independently. Under dwindling food security situation worldwide, it is being emphasized to go for advancement of local food production via increasing cropping intensity; meanwhile, ecological concerns are of serious issues (Kumar et al. 2002; Ali and Sarker, 2013). Modeling approach is reported to be an alternative tool in examining the competition and facilitation of different crops in relay cropping systems (Knorzer et al. 2011). Models for cereal–legume relay intercropping systems are extensively studied, but the system still bears the potency for more sustainable farming (Waddington et al. 1990; Whitmore and Schröder, 2007). While conducting the fertilizer management studies especially N, models could also be very helpful in predicting and assessing the fertilizer doses of various component crops in relay cropping systems, based on the information pertaining to their morphology, physiology, and phenology. In intercropping-based studies, though few models and submodels are pretty advanced in simulating interspecific and intraspecific competition for N or photosynthetically active radiation (PAR) (Munz et al. 2014), the development of more comprehensive models of relayed crops is still inevitable. To date, these models just assess the competition effects and stimulate its effects in terms of grain yield, biological yield, and harvest index; however, knowledge pertaining to yield gain has not been explained by these models (Jones et al. 2003). While designing relay cropping systems, modeling could also be helpful in adjusting various crops. Hence, the full use of the value-added chain of a crop growth model offers the possibility of increasing the knowledge about plant growth including plant–soil, plant–atmosphere, and plant–management interactions more comprehensively (Jurik and Van 2004). It is important to develop, evaluate, and validate those models intensively and carefully, but modeling and simulation should aim to improve intercropping analysis and hence optimize intercropping systems in the field (Seran and Brintha, 2010). Furthermore, there is a need for dynamic modeling tools to evaluate how wide ranges of soil conditions, various weather sequences, and different management schemes modify the yield and environmental impact of multispecies systems.

Model calibration for simulating the effects of interspecific interactions are of significant importance in determining and estimating the productivity potential and dynamics of relay crops (Knörzer et al. 2011). Plant interaction models are categorized as being either empirical, providing only a depiction of the outcome of competition, or process-based, offering a representation of the physiological processes underlying plant growth (Vincent and Harja 2002; Manson et al. 2006). Empirical models could be more crucial in parameterization of relay crops; nonetheless, mechanistic models can also be employed to simulate the performance and response of individual crops based on focal plant–neighbor plant interactions (Malezieux et al. 2009).

Adoption and scope

The potential benefits of adopting relay cropping are summarized in Fig. 3. Relay cropping systems are adaptable almost in every ecological conditions and edaphic characteristics (Table 3); for instance, Gupta and Bhowmick (2005) reported that rice–lathyrus and rice–lentil relay cropping are very popular cropping systems in West Bengal. He also reported that both crops performed well under muddy land conditions of rice and recorded higher net return than did sole cultivation of rice. In cotton–wheat cropping systems, cotton delays the sowing of wheat; therefore, some farmers use broadcasting of wheat seed into standing water in previous crops (Khan and Khaliq 2005). Although reduced germination and poor stand establishment are the big challenges with broadcasting of wheat in standing water, this issue can be solved by opting different physiological approaches like seed priming or seed coating. In relay cropping systems, seed priming may boost up the germination rate of interseeded crops and may avoid seed death through suffocation, which may be caused due to

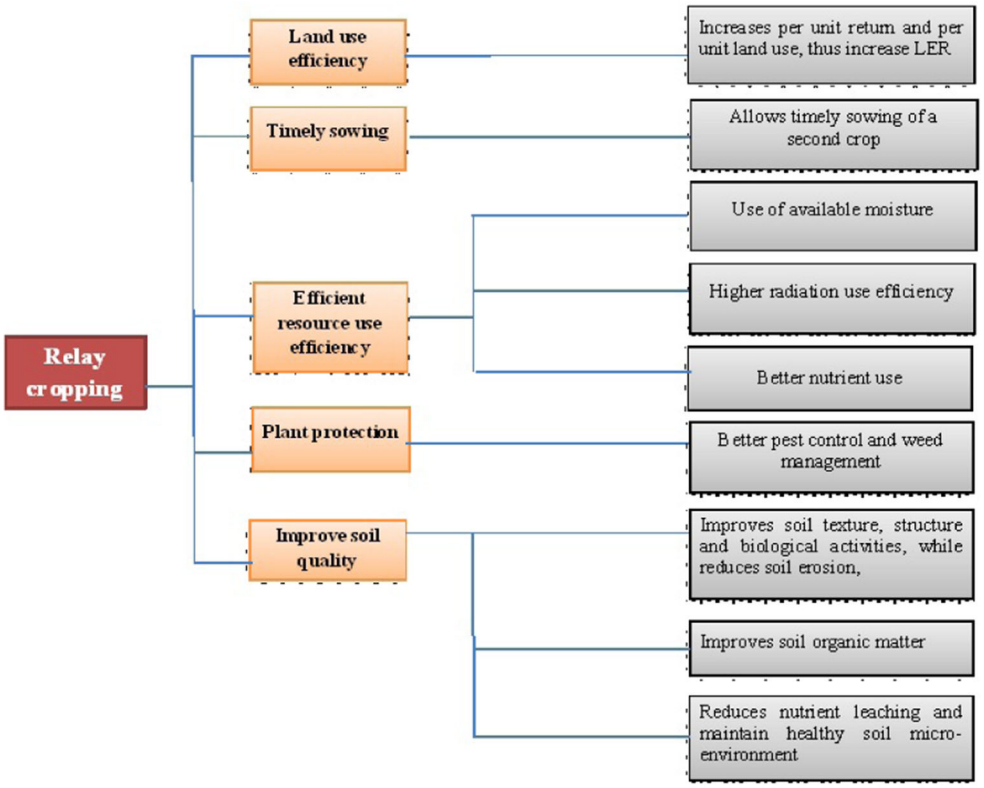


Fig. 3 Potential benefits of relay cropping

broadcasting of second crops in standing water. Seed treatment enhanced relayed soybean seedling growth and resistance to the lodging and also increased the seed yield under shading by corn (Yan et al., 2010). Furthermore, using above-normal seed rates that ensure desired plant stand can in turn help suppressing weed growth (Saeed et al. 2000). This situation warrants the optimization of both seed steeping duration and seeding density to harvest a successful crop of wheat relayed at zero tillage through surface seeding in the preceding sunflower crop (Saeed et al. 2000). Gupta and Bhowmick (2005) tested the effects of different seed priming times on the performance and growth of relayed lathyrus in rice and found that sowing of sprouted lathyrus and lentil seeds in rice-based relay cropping systems recorded more plant height, branches per plant, pods per plant, and seeds per pod of both crops, while soaking in water for 6 h gave less yield than did sprouted seed but more than did the non-primed dry seeds. In short, relay cropping can be an adaptable approach, but the issues of poor germination and stand establishment for certain crop species (when sown as relay crop) must be overcome to enhance the efficacy of a relay cropping system.

Conclusions

Relay cropping is a complex suite of new resource-efficient technologies, which possess the capability to improve soil, air, and water quality; increase net return and land equivalent ratio; and control weeds and pest infestation. Relay cropping compensates the yield loss of main crops by providing additional income from relayed crops. Furthermore, relay cropping maintains a natural ecosystem via crop diversification. There is also evidence for higher crop yields with relay cropping by the inclusion of legumes. However, more research is inevitable to further improve its adoption especially regarding the interaction of its key components with local soil and climate conditions. The present article synthesized the existing data to provide a comprehensive review on benefits, adaptation, and scope of relay cropping. Factors to be overcome in achieving and exploring the potential of relay cropping system include extensive research on all possible effects and dissemination of its production potential among farmers. Likewise, machinery development for sowing of relay crops, understanding response of succeeding crop to reduced tillage, changes in nutrient cycling, and nitrogen demand should be studied in the future. Demonstration trials on farmers' fields and technology transfer programs can hasten adoption of this system. On farm-research level, development and demonstration of the production potential of this system is required, including economic analysis and farm budgeting at regional and global levels. Extensive information has come from areas where this system has been practiced for some time, but it is at infant stages in many

Table 3 Opportunities of relay cropping in different cropping systems

Cropping system	Location of experiment	Sole crop	Candidate relay crop (s)	Problem before using relay cropping approach	Conclusive results/potential benefits	Reference
Organic grain production system	Southeastern France	Wheat	Four legume crops (*Medicago lupulina* L, *Medicago sativa* L, *Trifolium pretense* L and *Trifolium repens* L)	Nitrogen (N) management is a key issue in livestock-free organic grain systems.	N nutrition of the subsequent legumes after wheat induced a 30% increase in maize grain yield. This concluded that relay cropping of legume in winter wheat is good approach for improving N availability in organic grain production	Amossé et al. (2014)
Subsistence grain production system	Tigray in northern Ethiopia	Sorghum	Legume shrubs (*Sesbania sesban* L and *Cajanus cajan* L)	Lower soil fertility Less grain production	Relay cropping of legumes in sorghum increased sorghum grain yield by 63 to 130%, respectively, under non-fertilized conditions.	Reda et al. (2005)
Cotton–wheat cropping system	Bathinda, India	Cotton	Wheat	Delayed sowing of wheat with subsequent wheat yield reduction	Higher wheat grain yield (24.9%) and net return (29.2%) due to relaying of wheat in cotton	Buttar et al. (2011)
Wheat–potato cropping system	Shimla, India	Wheat	Potato	Delayed sowing of potato (*Solanum tuberosum* L.)	Higher tuber yield (38.3%) was observed.	Dua et al. (2007)
Upland rice cropping system	Bouaké, Côte d'Ivoire	Upland Rice	*Crotalaria juncea* L, *Cajanus cajan* L, *Mucuna pruriens* L. *Calopogoniummucunoides* L, and *Stylosanthes hamate* L	Decline in soil fertility due to intensive rice cultivation	The rice grain yield can be increased by relaying legumes in upland rice.	Akanvou et al. (2002)
Wheat-based cropping system	Anyang city, Henan province, China.	Wheat	Cotton	Delayed planting of cotton resulted in delayed and lower boll production in cotton.	Cotton produced more bolls and developed more efficiently yield-related traits. Therefore, relay cropping solves the conflict of timely sowing giving higher yield than sequential cropping systems do.	Zhang et al. (2008a,b)
Wheat-based cropping system/rye-based cropping system	Canada	Wheat/rye	Italian ryegrass and red clover	Soil erosion Decline in soil physical characteristics	These relay cropped forage crops provided 5.5% more ground cover thus reducing soil erosion. Conclusively, inclusion of forage relay crops in wheat or rye reduced soil erosion and thus increased soil physical characteristics	Edwards (1998)
Rain-fed lowland rice production system	Northeast Thailand	Lowland rice	*Stylosanthes guianensis* (Stylo)	Less rice production due to rain fed area and low soil fertility Shortage of forage for livestock	Relay cropping of stylo increase rice production by increasing soil fertility, and an aditional amount of stylo 367 g m^{-2} was obtained. Thus, a rice–stylo relay intercropping system is worth considering as one way to utilize paddy fields during dry seasons	Homma et al. (2008)
Forage production system	Northern Great plains of USA	Sorghum and maize	Camelina	Shorter growing seasons	Camelina-based relay treatment showed good potential for biofuel and energy feedstock and forage production in the northern Great Plains	Berti et al. (2015)

other areas receiving high rainfall and having suitable land for pastures. In the future, global food security and environmental preservation, together with farm-level profitability, should be the main focus for sustainable food production.

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